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Problem Set 4

I. CURRENCY RISK

Read the January 31 Buttonwood column in *The Economist,* "A peg in a poke."

1. In a few short lines, summarize what you think Philip Coggan (the columnist) thinks about the effects of currency pegs on currency volatility and exchange rate risk more generally.

   Hint: You may wish to make reference to "heavy-tailed" distributions or draw a density function to describe how currency volatility differs when there is a peg and when there is not.

2. Suppose you agree with Coggan's view and wish to speculate on the risks in pegged currencies (i.e. you think that his view is not fully reflected in market prices). Discuss an instrument you could use to take a position on your view.

   Hint: Compare the payoffs of the straddle and the strangle and discuss which is better suited to longing the risks in pegged currencies.

II. The Binomial Tree

1. Consider a binomial model with two equally likely states u and d "tomorrow" at Date T and two risky assets Asset 1, S^1, and Asset 2, S^2. Asset 1 pays off 100 in state u and zero in state d, i.e. $S^1_u = 100$ and $S^1_d = 0$. Asset 2 pays off zero in state u and 100 in state d, i.e. $S^2_u = 0$ and $S^2_d = 100$. S^1 has price 50 today, i.e. $S^1_0 = 50$, and S^2 has price 25 "today" at Date 0, i.e. $S^2_0 = 25$.

   a) Write down the Date-T payoffs of a European put option on Asset 1 with strike $K = 80$. Call the state-u payoff p_u and the state-d payoff p_d.

b) Find a replicating portfolio containing only Asset 1 and Asset 2 to replicate this put. Call the quantity of Asset 1 in the portfolio Δ_1 and the quantity of Asset 2 in the portfolio Δ_2.

c) Use the portfolio you found above to find the price p_0 of the put at Date 0. What theorem are you using?

d) Now suppose the put is American (it is still written on Asset 1 with strike $K = 80$). Write down the payoff of the put if it were to mature at Date 0, i.e. write down the *intrinsic value* of the option.

e) Given your answers above, do you want to exercise the American put at Date 0, or hold it till Date T?

f) What does your answer above say about how the potentiality value of the option compares to the deferred payment value?

2. Consider a binomial model with two securities, a risk-free bond and a risky stock. The risk-free rate is zero. The stock price is 100 today. The stock pays off 120 in state u and 90 in state d. State u and state d are equally likely.

   a) What is the discounted expected payoff of the stock? I.e. compute

   $$\frac{1}{1+r}\mathbb{E}\big[S_T\big].$$

   In the CAPM world, is the stock's beta positive or negative?

b) Write down the risk-neutral probabilities.

c) What is the value of a European call option with a strike price of 100?

What is the discounted expected payoff of the call? I.e. compute

$$\frac{1}{1+r}\mathbb{E}[c_T].$$

In the CAPM world, is the call's beta positive or negative?

d) What is the value of a European put option with a strike price of 100?

What is the discounted expected payoff of the put? I.e. compute

$$\frac{1}{1+r}\mathbb{E}[p_T].$$

In the CAPM world, is the put's beta positive or negative?

e) Verify that the European call and European put satisfy put-call parity.